DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) • PROFESSIONAL DATA ANALYST TRACK

Job Market Intelligence Solution

An end-to-end data pipeline transforming raw job postings into evidence-based career insights.

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PROBLEM & OBJECTIVES

Bridging the Gap in Job Market Transparency

The Challenge

Job seekers struggle to identify true demand due to fragmented platforms and inconsistent titles.

The Solution

A clean, documented, and reproducible analytical workflow converting 17-field datasets into insights.

Primary Audience

Targeting job seekers, career changers, and training providers with evidence-backed guidance.

ARCHITECTURE

Medallion Pattern & Star Schema Warehouse

Layer 01

Bronze Layer

Ingests raw files unchanged for complete traceability and auditability.

Layer 02

Silver Layer

Cleans and standardizes titles, locations, and validates flags while keeping originals.

Layer 03

Gold Layer

Star schema with fact postings, dimensions, and skill bridge table for analytics.

MARKET DEMAND

Role Volume & Geographic Distribution

17

Core Fields Profiled & Audited

Role Family Identification

Categorizes job postings into clear functional groups for comparison.

Platform Source Mix

Evaluates job board origins (`job_via`) and posting frequency patterns.

SKILLS INTELLIGENCE

Requested Competencies & Co-Occurrence

Skill Parsing

Converts `job_skills` into normalized job-skill pairs without inflating posting counts.

Co-Occurrence Analysis

Identifies core tool combinations frequently requested together across specific roles.

Skill Types

Distinguishes technical tools from programming languages and domain proficiencies.

SALARY ANALYSIS

Annual & Hourly Compensation Transparency

Separate Pay Metrics

Keeps `salary_year_avg` and `salary_hour_avg` strictly separated and guided by `salary_rate`.

Robust Statistics

Reports medians and interquartile ranges alongside explicit disclosure sample sizes.

Disclosure Impact

Evaluates selective salary reporting share and its effect on market reliability.

FORECASTING

Predictive Modeling & Trend Validation

Volume Forecasting

Evaluates monthly posting volume models against a seasonal-naive baseline using MAE & RMSE.

Regression Analysis

Exploratory regression modeling of annual salary against role and work attributes with holdout validation.

Temporal Aggregation

Builds weekly and monthly aggregates from `job_posted_date` with comparable period checks.

BI DASHBOARDS

Tableau & Power BI Unified Delivery

6-Page Structure

Executive Overview, Job Demand, Skills Intelligence, Salary Analysis, Work Attributes, and Methodology.

Power BI Semantic Model

Configured with Power Query transformations and documented DAX measures.

Cross-Validation

Independent reconciliation between SQL, Python, Tableau, and Power BI totals.

OUTCOMES

Evidence-Backed Recommendations & Caveats

For Career Seekers

Focus upskilling on verified high-demand skill pairs and high-disclosure role sectors.

For Training Providers

Align curricula with empirical co-occurrence patterns rather than isolated tool requirements.

Stated Limitations

Postings measure advertised demand, not actual hires; salary figures reflect selective disclosure.

Thank You

Open floor for questions and live dashboard demonstration.

Digital Egypt Pioneers Initiative (DEPI) • Job Market Analysis Team